

Question number	Answer	Additional guidance	Mark
(a)(i)	An explanation that includes three of the following points: • less calcium ions would bind to troponin (1) • troponin doesn't change shape (1) • (therefore) tropomyosin remain in position / less tropomyosin moves (1) • myosin head cannot bind to {actin / binding site} / {no / less} formation of (actin-myosin) cross bridges} (1) • (therefore) muscles stay relaxed	ACCEPT tropomyosin blocks binding sites / binding sites not exposed ACCEPT {less / no} {contraction of muscle / shortening of sarcomere} ignore disrupt contraction	(3)

Question number	Answer	Additional guidance	Mark
(b)(i)	An answer that includes the following point: homeostasis / osmoregulation / negative feedback (1)	ACCEPT phonetic spelling	(1)

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(b)(ii)	A calculation showing the following steps: - calculation of (upper and /or lower) range of serum calcium (1) - correct calculation of difference (1)	$(91 - 6\%) = 85.54$ OR $(91 + 6\%) = 96.46$ accept 12% / 5.46 / 5.5 (mg dm^{-3}) for 1 mark $(96.46 - 85.54) = 10.92 / 10.9$ (mg dm^{-3}) Accept 11 (mg dm^{-3}) Accept answers in standard form $1.09 \times 10^1 / 1.1 \times 10^1$ correct answer gains both marks	(2)

Question number	Answer	Additional guidance	Mark
(c)(i)	An explanation that includes two of the following points: • (small) increase in lactate concentration (between 15 to 60 minutes / during exercise) / little change in lactate concentration (between 15 to 60 minutes) (1) • because {lactate / lactic acid} is produced by anaerobic respiration (1) • lactate produced is being converted into pyruvate (1)	ACCEPT increases, then decreases then increases again / fluctuates ignore if refer to change between 0 – 60 mins lactate concentration in blood stays high as equilibrium between {lactate going to the liver / being converted to pyruvate} and continued anaerobic respiration during exercise	(2)

Question number	Answer	Additional guidance	Mark
(c)(ii)	An answer that includes one of the following points: • to provide {base line data / starting point / for comparison} (1)	to provide lactate concentration when {anaerobic respiration is not occurring / at rest} identify if exercise affects blood lactate concentration ignore control	(1)

Question number	Answer	Additional guidance	Mark
(c)(iii)	An answer that includes the following point: a named controlled variable with a suitable description on how it could be controlled (1)	e.g. same age - select any age range eg 18 – 35 yrs / same age Same (resting) blood pressure Same VO₂ max Same cycling intensity by doing same course on exercise bike Same exercise bike Same BMI Same {food / fluid} intake (24hours) prior to exercise – standardized meals same {caffeine / drug} intake same {temperature / humidity / oxygen concentration} of room etc ignore similar	(1)

